

Lab 3

Relational & Logical operators and conditional

Learning Objectives

The following types will be covered in this lab session

- To understand relational operators in C++.
- To learn logical operators and their use in decision making.
- To implement conditional statements (if, if-else, nested if).
- To solve problems using conditions in programming.

Relational Operators

Relational operators are used to compare two values. The result of a relational operation is either true (1) **or** false (0). These operators are mainly used in decision-making and loops.

They help the program decide conditions like whether one value is greater, smaller, or equal to another.

Syntax:

```
a == b // equal to
a != b // not equal to
a > b // greater than
a < b // less than
a >= b // greater than or equal to
a <= b // less than or equal to
```

Example:

```
#include <iostream>
using namespace std;
int main() {
    int a = 10, b = 5; // Declare variables
    cout << (a > b) << endl; // 1 (true)
    cout << (a < b) << endl; // 0 (false)
    cout << (a == b) << endl; // 0 (false)
    return 0; }
```

2. Logical Operators

logical operators are used to combine multiple conditions. They are used when we need to check more than one condition at a time in decision-making.

They return either true or false based on combined conditions.

Syntax:

```
&& // AND (both conditions must be true)
```

```
|| // OR (any one condition true)
```

```
! // NOT (reverses condition)
```

Example:

```
#include <iostream>
using namespace std;
int main() {
    int a = 10, b = 5;
    cout << (a > 5 && b < 10) << endl; // 1 (true)
    cout << (a < 5 || b < 10) << endl; // 1 (true)
    cout << !(a > b) << endl;          // 0 (false)
    return 0;
}
```

3. Conditional Statements (if, if-else, nested if)

if Statement:

The if statement is used to execute a block of code only when a condition is true. If the condition is false, the block is skipped

Syntax:

```
if (condition) {  
    // code executes if condition is true  
}
```

Example:

```
#include <iostream>  
using namespace std;  
int main() {  
    int num = 10;  
    if (num > 0) {  
        cout << "Number is positive" << endl;  
    }  
    return 0;  
}
```

If-else- Statement:

The if-else statement is used when we have two choices. If the condition is true, one block executes; otherwise, the else block executes.

Syntax:

```
if (condition) {  
    // true block  
} else {  
    // false block  
}
```

Example:

```
#include <iostream>  
using namespace std;  
int main() {  
    int num;  
    cout << "Enter a number: ";  
    cin >> num;  
    if (num % 2 == 0) {  
        cout << "Even number" << endl;  
    } else {  
        cout << "Odd number" << endl;  
    }  
    return 0;  
}
```

Nested If Statement:

A nested if is an if statement inside another if statement. It is used when multiple conditions depend on each other.

Syntax:

```
if (condition1) {  
  
    if (condition2) {  
        // code  
    }  
}
```

Example:

```
#include <iostream>  
using namespace std;  
  
int main() {  
  
    int age = 20;  
  
    if (age >= 18) {    //check if the condition true  
  
        if (age <= 60) { //check if the 2nd condition true  
            cout << "Adult" << endl;  
        }  
    }  
    return 0;}
```

In this lab, we studied relational operators, logical operators, and conditional statements in C++. We learned how relational operators compare values, how logical operators combine conditions, and how conditional statements help in decision-making. These concepts are essential for building logical and real-world programs.

Lab 3: Tasks

Task 1: Comparison / Relational Operators

Use comparison operators to check if the first number is equal to, greater than, or less than the second number.

Display appropriate messages based on the comparison results.

```
#include <iostream>           // Header file for input/output
using namespace std;         // Standard namespace
int main(){                  // Main function
    int num1, num2;          // Variables for two numbers
    cout << "Enter first number: "; // Ask user
    cin >> num1;              // Input first number
    cout << "Enter second number: "; // Ask user
    cin >> num2;              // Input second number
    if(num1 == num2)         // Compare equal
        cout << "First number is equal to second number" : // Output
    if(num1 < num2)           // Compare less
        cout << "First number is less than second number" << endl;
    return 0;                // End program
}
```

Task 2: Logical Operators

Write a C program that prompts the user to enter an integer.

Use logical operators to check if the entered number is greater than 0 and less than 100.

Display a message indicating whether the number meets both conditions or not.

```
#include <iostream>                // Header file

using namespace std;              // Standard namespace

int main() {                      // Main function

    int a, b;                     // Variables

    cout << "Enter first number: "; // Ask user

    cin >> a;                     // Input

    cout << "Enter second number: "; // Ask user

    cin >> b;                     // Input

    if(a > 0 && b > 0)            // AND operator

        cout << "Both numbers are positive" << endl; // Output

    if(a > 0 || b > 0)            // OR operator

        cout << "At least one number is positive" << endl;

    if(!(a == b))                 // NOT operator

        cout << "Numbers are not equal" << endl;

    return 0;                    // End
```

Task 3: If Statement

Write a C program that prompts the user to enter an integer.

Use an if statement to check if the entered number is positive.

Display a message indicating whether the number is positive or not.

```
#include <iostream>           // Header file

using namespace std;         // Standard namespace

int main()                   // Main function
{
    int num;                  // Variable

    cout << "Enter a number: "; // Ask user

    cin >> num;                // Input

    if(num > 0)                // Check condition

        cout << "Number is positive" << endl; // Output

    return 0;                 // End

}
```

Task 4: If-Else Statement

Write a C program that prompts the user to enter their age.

Use an if-else statement to check if the entered age is greater than or equal to

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```
#include <iostream>           // Header file

using namespace std;         // Standard namespace

int main()                   // Main function

    int num;                 // Variable

    cout << "Enter a number: "; // Ask user

                                // Input
    cin >> num;

    if(num % 2 == 0)          // Check even

        cout << "Number is even" << endl; // Output
    else                      // Otherwise

        cout << "Number is odd" << endl;

    return 0;                 // End

}
```

Task 5: Else-If Statement

Write a C program that prompts the user to enter an integer.

Use an else-if statement to check if the entered number is positive, negative, or zero.

Display appropriate messages based on the entered number.

```
#include <iostream>                // Header file

using namespace std;              // Standard namespace

int main(){                       // Main function

    int marks;                   // Variable

    cout << "Enter marks: ";     // Ask user

    cin >> marks;                // Input

    if(marks >= 80)               // Condition 1

        cout << "Grade A" << endl; // Output

    else if(marks >= 60)          // Condition 2
```

```
    cout << "Grade B" << endl;    // Output

    else if(marks >= 40)            // Condition 3

        cout << "Grade C" << endl;    // Output

    else                            // Otherwise

        cout << "Fail" << endl;        // Output

    return 0;                        // End

}
```

Task 6: Nested If Statement

```
#include <iostream>                // Header file

using namespace std;               // Standard namespace

int main()                          // Main function
{

    int age;                        // Variable for age

    char nationality;               // Variable for nationality

    cout << "Enter age: ";          // Ask user

    cin >> age;                     // Input age

    cout << "Enter nationality (P for Pakistani): "; // Ask user
```

```
cin >> nationality;           // Input character

if(age >= 18) {                // Outer if
    if(nationality == 'P')      // Inner if

        cout << "Eligible for CNIC" << endl; // Output

    else                        // Inner else

        cout << "Not Pakistani" << endl;

}

else                            // Outer else

    cout << "Under age" << endl;

return 0;                       // Ended Successfully
}
```